KARTHIK SAI S

AI/ML Engineer | Python | Machine Learning | Generative AI | Agentic AI | Computer Vision | LLMs | RAG
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PROFESSIONAL SUMMARY

AI/ML Engineer with 6+ months of internship experience building Generative AI, RAG, NLP and Computer Vision systems, plus 3 months of ASP.NET Core backend engineering. Built a Text-to-SQL system (85% pass rate on a 40-case benchmark), a DeBERTa-v3 email classifier (~96% accuracy) and a YOLO11 traffic-violation detector. Skilled in agentic AI and LLM fine-tuning. Open to AI, GenAI and ML roles in Chennai, Bengaluru, Hyderabad, Kochi and Coimbatore.

TECHNICAL SKILLS

Languages & Data:	Python, SQL, T-SQL, C#, pandas, NumPy
Machine Learning:	scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, NLP, Feature Engineering, Model Evaluation, Class Imbalance Handling, Data Leakage Prevention, Learning-to-Rank, SHAP
Computer Vision:	YOLO11 (Ultralytics), Object Detection, OpenCV, Data Augmentation, PaddleOCR
Generative AI & LLMs:	LLMs, RAG, Prompt Engineering, LLM Fine-Tuning (LoRA / QLoRA, PEFT), Embeddings, ChromaDB, Hybrid Search (BM25 + Dense), Reranking, LLM Evaluation, Guardrails
Agentic AI:	Multi-Agent Systems, LangGraph, MCP, Tool Calling, Human-in-the-Loop Workflows
Backend & Databases:	FastAPI, REST, gRPC, ASP.NET Core, Dapper, JWT, OAuth 2.0, SQL Server, PostgreSQL
Cloud & Tools:	Git, GitHub, Google Cloud Platform (GCP), Google Cloud Pub/Sub, Gmail API

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern | WeServeCodes, Ahmedabad, India — Apr 2026 - Present

- Built Text-to-SQL, email-routing and ATMS traffic-violation systems (see Projects), exposed as FastAPI services with confidence routing, AST-based SQL validation and PII guardrails.
- Built a read-only AI support chatbot (Groq LLM, sqlglot Query Guard, PaddleOCR) that generates fix SQL and step-by-step instructions for support-team data issues.
- Built a leakage-safe supplier ranking advisor: LambdaMART raised NDCG@3 from 0.758 to 0.836, plus price (R2 0.84) and return-risk (AUC 0.76) sub-models with TreeSHAP explanations.

.NET Backend Developer Intern | WeServeCodes, Ahmedabad, India — Jan 2026 - Mar 2026

- Developed REST and gRPC services in ASP.NET Core (Clean Architecture, CQRS with MediatR) for a production SQL Server application.
- Migrated ADO.NET data access to Dapper, fixed N+1 query issues, and added JWT authentication, rate limiting and idempotency safeguards.

PROJECTS

ATMS: AI Traffic Violation Detection for Highway CCTV — Live Production Project

Python, YOLO11 (Ultralytics), PyTorch, OpenCV, RTSP, FastAPI

- Fine-tuning YOLO11 on a curated multi-source dataset of 15,000+ annotated images to detect helmet, no-helmet, triple riding, seatbelt and vehicle classes from live RTSP highway cameras.
- Designed a two-stage pipeline (vehicle detection, then cropped violation detection) with a head-count rule for triple riding, handling class imbalance with oversampling and targeted augmentation.
- Used time-block data splits to prevent leakage and evaluated per class on a held-out test set against precision/recall targets aligned to NHAI benchmarks.

Enterprise Text-to-SQL / AI Database Assistant — github.com/KarthikSai08/enterprise-ai-assistant

Python, FastAPI, Groq (LLaMA 3.1-8B), Ollama, BGE-M3, BM25, ChromaDB, BGE-Reranker-v2-m3, sqlglot, SQL Server

- Designed two-stage hybrid retrieval (BM25 + BGE-M3 + Reciprocal Rank Fusion) with cross-encoder reranking to pick the top 2-5 tables from a 68-table, 717-column schema, with BFS join-graph traversal across 149 foreign keys.
- Built a three-layer hallucination defense (retrieval-constrained context, sqlglot AST validation, corrective retries) with 30 PII guardrail patterns; 85% pass rate on a 40-case benchmark, validated on 265 manual questions.

AI-Powered Email Classification & Real-Time Routing — github.com/KarthikSai08/ai-email-automation

Python, PyTorch, Hugging Face Transformers (DeBERTa-v3), FastAPI, Gmail API, OAuth 2.0, Google Cloud Pub/Sub

- Fine-tuned DeBERTa-v3-base for 10-class department classification (~96% accuracy); low-confidence mail below 0.70 goes to a default address.
- Built an event-driven pipeline (Gmail API watch, Pub/Sub push) that persists message and history IDs for exactly-once processing across restarts.

EDUCATION

Master of Science in Computer Science | Hindustan College of Arts and Science — 2025 | CGPA: 8.0/10
Bachelor of Science in Computer Science | Dhanraj Baid Jain College of Arts and Science — 2023 | CGPA: 7.9/10

PUBLICATIONS

A Review on Supervised Machine Learning Algorithms - ICFTAI'24 (Intl. Conf. on Future Trends in AI), Oct 2024